INCH-POUND

A-A-54761
20 November 1992

COMMERCIAL ITEM DESCRIPTION

TIP ASSEMBLY, SONIC PROPHYLAXIS UNIT, DENTAL

The General Services Administration has authorized the use of this Commercial Item Description.

This commercial item description covers tip assemblies used with an air driven, sonic prophylaxis unit, used for removing calculus.

Salient characteristics:

Types

Type I - Universal
Type II - Sickle

The Type I shall be a universal tip suitable for all surfaces for removal of both heavy and medium calculus and stain from tooth surfaces.

The Type II shall be a sickle tip suitable for use on sensitive patients and provide alternate access for difficult posterior areas for removal of calculus and stain.

The tips shall be designed for use with sonic prophylaxis units (See note 2).

Each tip shall be furnished with a pliable clear or translucent cover for protection and storage. The cover shall be capable of being cold sterilized. The tip assembly shall properly screw into the handpiece.

The style, design and dimensions of the tips shall conform to Defense Personnel Support Center (DPSC) Drawing No. 23401. When all the components are assembled with the handpiece, the tip assembly and the hollow shaft which extends the length of the handpiece shall form a rigid connection.

AMSC N/A FSC 6520

DISTRIBUTION STATEMENT A. Approved for public release; distribution is unlimited.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T09:38Z
Check the source to verify that this is the current version before use.

Material.

Tip assemblies - operative end	Martensitic stainless steel, Type 440C.
Holder	Anodized aluminum alloy (6061-T6).
Base	Martensitic stainless steel, Type 416.
Clean-out wire	Austenitic stainless steel.

The tips shall be self-tuning and shall vibrate (resonate) at a frequency of 6100 plus or minus 300 Hz with the universal tip amplitude of 0.010 inch, plus 0.001 inch, minus 0.003 inch, peak to peak, orbital motion.

Water shall be delivered to the tip by passing through a small integral tube within the tip holder. The design and vibration characteristics of the tips shall allow water to flow by surface contact effect to within $^{1}/_{4}$ inch of the free end of the tip before spraying occurs.

The water channel of the tip assemblies (tips and tip holders) shall be "office cleanable" with the use of clean-out wires to clear any blockage of the water channel. The tip assemblies shall be autoclavable, dry heat sterilizable, and quickly changeable. The tips shall have a diamond cross-section and shall incorporate a spherical tip to give clinical function with minimum tooth damage and maximum patient comfort.

Hardness. After proper stress relieving, the Rockwell hardness of the operative ends shall be C56 to C60, when tested in accordance with ASTM E18 or ASTM E92 and E140.

Passivation. All stainless steel shall be properly passivated.

Corrosion resistance. The tips shall show no signs of corrosion when tested in accordance with the copper sulfate and boil test requirements of ASTM F1089.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T09:38Z
Check the source to verify that this is the current version before use.

Autoclavability. Tip assemblies (tip and tip holder), shall be capable of being autoclaved without showing any signs of corrosion, reduction in performance, or malfunction.

Operational test. The tip assembly shall be attached to the handpiece. The unit shall be operated, "run-in", for 4 hours under normal operating conditions. The tip shall then be inspected and tested for compliance with the specification requirements.

Finish. The operative ends of the tips shall have a smooth bright finish except for the sharpened edges which shall have a finely honed finish. The operative ends shall have sharp cutting edges and shall be free of defects such as coarse grind marks, cracks, gouges, burrs, feathers, or other imperfections which affect performance.

Color coding. The holder of each tip assembly shall be color coded as follows: Universal-black, sickel-blue.

Workmanship. The tip assembly shall be free from defects which detract from its appearance or impair its serviceability.

Unit. Each (EA). One tip assembly of the type specified, constitutes one unit.

Quality Assurance Provisions.

Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

Records. Records of examinations and tests performed by or for the contractor shall be maintained by the contractor and made available to the Government upon the Government's request, at any time, or from time to time, during the performance of the

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T09:38Z
Check the source to verify that this is the current version before use.

Records. Records of examinations and tests performed by or for the contractor shall be maintained by the contractor and made available to the Government upon the Government's request, at any time, or from time to time, during the performance of the contract and for a period of three years after the delivery of the supplies to which such records relate.

Inspection. Inspection, as used herein, is defined as both examination (such as visual or auditory investigation without the use of special laboratory appliances or procedures) and testing (determination by technical means of physical and chemical properties) of the item.

Examination. The tips shall be examined to determine compliance with specification requirements.

Sampling for examination. Sampling for examination shall be conducted in accordance with MIL-STD-105 and as specified herein. Unit of product for examination purposes shall be one tip as specified. Sampling shall be inspection level II with an Acceptable Quality Level (AQL) of 1.0.

Tests. Tests shall be conducted to determine compliance with specification requirements.

Sampling for tests. Sampling for tests shall be conducted in accordance with MIL-STD-105 and as specified herein. Unit of product for test purposes shall be one tip. Sampling shall be inspection level S-2 with an AQL of 1.0.

Autoclave test. The tips shall be wrapped in muslin and autoclaved at 270 degrees F plus or minus 5 degrees F and 27 plus or minus 1 pound per square inch of steam for a minimum of 20 minutes. The tip shall then be inspected and tested for compliance with specification requirements.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T09:38Z
Check the source to verify that this is the current version before use.

Metric products. Products manufactured to metric dimensions will be considered on an equal basis with those manufactured using inch-pound units, providing they fall within the tolerances specified using conversion tables contained in the latest revision of Federal Standard No. 376, and all other requirements of this document are met.

If a product is manufactured to metric dimensions and those dimensions exceed the tolerances specified in the inch/pound units, a request should be made to the contracting officer to determine if the product is acceptable.

The contracting officer has the option of accepting or rejecting the product.

Contractor certification. The contractor shall certify and maintain substantiating evidence that the product offered meets the salient characteristics of this Commercial Item Description, and that the product conforms to the producers' own drawings, specifications, standards, and quality assurance practices. The government reserves the right to require proof of such conformance prior to first delivery and thereafter as may be otherwise provided for under the provisions of the contract.

Regulatory requirements.

Federal Food, Drug and Cosmetic Act. If the product covered by this document has been determined by the U.S. Food and Drug Administration to be under its jurisdiction, the offeror/contractor shall comply, and be responsible for compliance by its subcontractors/suppliers, with the requirements of the Federal Food, Drug and Cosmetic Act, as amended, and regulations promulgated thereunder. In addition, the offeror/contractor shall comply, and be responsible for compliance by its subcontractors/suppliers, with the requirements of all other applicable Federal, State, and local statutes, ordinances, and regulations.

Recovered materials. The offeror/contractor is encouraged to use recovered material in accordance with Federal Acquisition Regulation Subpart 23.4 to the maximum extent practical.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T09:38Z
Check the source to verify that this is the current version before use.

Preservation, packaging, packing, labeling, and marking. Unless otherwise specified, preservation, packaging and packing shall be to a degree of protection to preclude damage to containers and/or contents thereof under normal shipping conditions, handling, etc., involving shipment from the supply source to the receiving activity, plus reshipment from receiving activity, and shall conform to applicable carrier's rules and regulations. Intermediate and exterior package quantities and labeling and marking shall be as specified in the contract and/or order.

Ordering data. Intermediate/exterior package quantities, labeling, and marking must be as specified in the contract and/or order.

NOTE (1): The following National Stock Numbers are covered by this document:

National Stock Numbers	Item Identifications
6520-01-210-4447	TIP ASSEMBLY, SONIC PROPHYLAXIS UNIT, DENTAL, Universal, Type I
6520-01-246-0199	TIP ASSEMBLY, SONIC PROPHYLAXIS UNIT, DENTAL, Sickle, Type II

NOTE (2): The following National Stock Numbers are referenced by this document:

National Stock Numbers	Item Identifications
6520-01-162-2906	SONIC PROPHYLAXIS UNIT, Dental Portable, Air Driven
6520-01-204-8688	SONIC PROPHYLAXIS UNIT, Dental Portable, Air Driven

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T09:38Z
Check the source to verify that this is the current version before use.

MILITARY INTERESTS:

Custodians:

Army-MD
Navy-MS
Air Force-03

Preparing Activity:

DoD-MB

Agent:

DLA-DM

CIVIL AGENCY COORDINATING ACTIVITIES:

VA-OSS
USPHS
FDA-MPQAS

Project No. 6520-2523
Location: ML Disk #2
(SONIC)

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T09:38Z
Check the source to verify that this is the current version before use.

STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

INSTRUCTIONS

1. The preparing activity must complete blocks 1, 2, 3, and 8. In block 1, both the document number and revision letter should be given.
2. The submitter of this form must complete blocks 4, 5, 6, and 7.
3. The preparing activity must provide a reply within 30 days from receipt of the form.

NOTE: This form may not be used to request copies of documents, nor to request waivers, or clarification of requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements.

I RECOMMEND A CHANGE:	1. DOCUMENT NUMBER	2. DOCUMENT DATE (YYMMDD)
	A-A-54761	20 November 1992

3. DOCUMENT TITLE
TIP ASSEMBLY, SONIC PROPHYLAXIS UNIT, DENTAL

4. NATURE OF CHANGE *(Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets as needed.)*

5. REASON FOR RECOMMENDATION

6. SUBMITTER

a. NAME (Last, First, Middle Initial)	b. ORGANIZATION
c. ADDRESS (Include Zip Code)	d. TELEPHONE (Include Area Code) (1) Commercial (2) AUTOVON (If applicable)

7. DATE SUBMITTED (YYMMDD)

8. PREPARING ACTIVITY

a. NAME	b. TELEPHONE (Include Area Code)
Defense Personnel Support Center ATTN: DPSC-MSE	(1) Commercial: (215)7372870 (2) AUTOVON: 444-2870
c. ADDRESS (Include Zip Code) 2800 South 20th Street Philadelphia, PA 19145	IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT: Defense Quality and Standardization Office 5203 Leesburg Pike, Suite 1403, Falls Church, VA 22041-3466 Telephone (703) 756-2340 AUTOVON 289-2340

DD Form 1426, OCT 89 — *Previous editions are obsolete*

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T09:38Z
Check the source to verify that this is the current version before use.